EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

Library Book Issue Request Form (Administration & Security Use Case)

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1. Aim

To plan and apply security features, role-based access control (RBAC), data permissions, safe scripting, and auditing practices in a Zoho Creator application (Library Book Issue Request Form) to ensure data protection, compliance, and controlled administrative oversight.

The experiment covers user management, profile configuration, record-level and field-level permissions, secure Deluge scripting, audit logging, backup strategies, and platform governance.

2. Procedure

Step 1: Define security and governance requirements. Identified sensitive data fields (student details, book status, fine records) and determined access levels for students, librarians, and system administrators.

Step 2: Configure role-based access control (RBAC). Created user profiles and roles in Zoho Creator to segment permissions between end-users and administrators.

Step 3: Set field-level and record-level permissions. Restricted visibility and edit permissions based on user login profiles and custom criteria rules.

Step 4: Implement secure scripting and validation. Ensured user inputs are sanitized, secure parameters are used in API calls, and exception handling is in place.

Step 5: Enable audit logs and system monitoring. Configured system audit trails to track record creation, modification, and deletion events.

Step 6: Plan backup and maintenance policies. Established routine data export schedules and administrator governance guidelines.

Note: This is a structured administrative implementation. Actual permission settings and governance policies follow platform configuration standards.

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3. Output

3.1 Security Matrix and Role Permissions

The security matrix outlines access rights across user profiles for the Library Book Issue application:

Profile	Book Request Form	Issue Approval	Fine Records	System Logs
Student	Create / View Own	No Access	View Own	No Access
Librarian	View All	Read/Write	Read/Write	View Basic
Administrator	Full Access	Full Access	Full Access	Full Access

3.2 Secure Scripting & Validation Practices

All Deluge scripts governing book issuance incorporate input sanitization, checking for valid student IDs and book availability prior to database insertion. Error handling blocks catch unexpected exceptions and log failure states without exposing sensitive stack traces to end users.

3.3 Audit Trails and Compliance

Zoho Creator's built-in audit logs were configured to record every administrative override, book return extension, and fine waiver. This ensures complete non-repudiation and accountability across all library transactions.

4. Final Result

Successfully planned and structured the implementation of robust security controls, granular role-based permissions, secure scripting practices, and administrative governance for the Library Book Issue Request Form application in Zoho Creator.

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